Jaden Dew

Berkeley, CA | jaden.dew1@gmail.com | (951) 440-9175 | LinkedIn | Personal Website

Education

University of California, Los Angeles Expected Graduation: June 2027
M.S. Aerospace Engineering

University of California, Berkeley May 2026
B.S. Aerospace Engineering, GPA: 3.843/4.0
Relevant Coursework: Intro to Compressible Flows, Combustion Processes, Heat Transfer

Skills

Technical: CFD, CAD, Lathe, Mill, Waterjet, Aerodynamics, Propulsion, Test Operations, Design Optimization
Software: Python, MATLAB, Julia, Siemens NX, Fusion 360, OpenFoam, Ansys, WinPlot, Git, Command Line
General: Communication, Leadership, Teamwork, Organization, Problem Solving

Work Experience

Vast Space | Propulsion Test Intern May 2025 – Aug 2025
Mojave, CA

- Conducted vacuum chamber hot-fire test operations of Impulse Space's Saiph thruster resulting in 24+ successful hot-fires, 9 successful test campaigns, and the completion of Haven Demo's flight rationale.
- Managed the flow of hot-fire data and conducted data reviews utilizing LabView and WinPlot diagnosing anomalies and assisting REs with developing fixes resulting in smooth test operation across 9 campaigns.
- Designed and constructed a COPV fill measurement and heating system in Siemens NX integrating it with a high-altitude test stand in support of holistic flight like testing of Haven 1's propulsion system.
- Designed mounting infrastructure for heat exchanger upgrade of a vacuum chamber test stand resulting in the successful installation of the upgrade.

NASA Ames Research Center | Aerodynamics Research Intern Jun 2024 - Dec 2024
Mountain View, CA

- Utilized DAFoam to develop 8+ reduced C_d airfoils for Titan's atmosphere across multiple AOAs, Reynolds, and Mach numbers, validating results in OpenFoam over 20+ AOAs at flight representative operating points.
- Developed a 3D shape optimization script in Julia, utilizing it to develop 2 FM enhanced rotor blade geometries resulting in a 2% overall increase in FM during hover confirmed in NASA OVERFLOW CFD.
- Presented results to NASA's Dragonfly team pitching this optimization pipeline for future rotor designs.

FLOW Lab – Dr. Simo Mäkiharju | Undergraduate Research Assistant Feb 2024 - May 2024
Berkeley, CA

- Developed surface roughness measurement procedure and MATLAB profile filtering and post-processing script to compute R_a, R_{ku}, R_{sk}, and sand grain roughness of 15+ superhydrophobic surfaces and coatings.

Project Experience

Space Enterprise at Berkeley | Propulsion Engineer Sep 2022 – May 2026

- Designed and fabricated the main valve assembly for a vertical propulsion system test-stand resulting in multiple successful cold-flows and 3+ successful hot-fire tests of Eureka-3's propulsion system.
- Own the design of a nitrous side custom poppet main valve for Eureka-3's flight propulsion system including CAD, FEA, and pressure loss sims resulting in multiple successful cold-flows.
- Owned the maintenance, assembly, and integration of the Lightbulb engine for the Eureka-2 liquid rocket including flight and ground testing operations resulting in 3 successful hot-fires and a successful launch.